

# Compact districting plans that better respect administrative boundaries

For California, Florida, New York, and Texas, we used a computer algorithm to find a substantially more compact (Cpt. = Polsby-Popper score) districting plans given a fixed number of times counties are divided between districts. Grey dots represent 5,000 simulated nonpartisan baseline plans.

## Takeaways

- In all four states, it is possible to create more compact plans that keep more counties together than in the enacted plans.
- These plans would also create Congressional delegations that (1) are substantially more competitive and (2) tend to more closely reflect statewide voting patterns.
- However, plans with the highest levels of compactness and respect for administrative boundaries do not necessarily produce proportional representation.

## How to interpret each plot

For example, in California it is possible to draw plans with 33% fewer county splits (27 → 18) that are also substantially more compact (0.21 → 0.35). Green labels indicate optimizations, while purple labels show the 2026 enacted plans. The statewide vote share shows the proportional allocation under the 2024 presidential election.

